G2F Phenotypic data 2023

The experimental design was a modified randomized complete block design, with the objective of balancing the need for within-site replication against the overall project goal of testing as many different hybrids as possible at each trial location. A common set of 30 check hybrids was used to connect environments (combinations of location and year), and the check hybrids were fully replicated in each location (family called “CHECK”). The remaining hybrids were replicated 0, 1 or 2 times, depending on seed availability.

The main group of hybrids evaluated in the main experiment 2023 (main experiment called “G2F_Main”) consists of doubled haploids that were testcrossed with ex-PVP inbred tester LH244. The doubled haploids were generated from F_1 hybrids resulting from crosses between two groups of parents. One group of parents consisted of inbreds LH212Ht, PHJ89, PHK76, PHN46, and PHP02, which are members of the non-stiff stalk heterotic group. The second group of parents consisted of GEMN-0096, GEMN-0097, GEMN-0192, and GEMN-0225, which are populations released by the Germplasm Enhancement of Maize (GEM) project.

Additional smaller-scale experiments were conducted alongside the main experiment for additional phenotyping and/or deployment of novel phenotyping methods. These experiments included the external Yellow Stripe experiment (experiment called “YS”), which involved a set of hybrids know as ‘Yellow Stripe’ hybrids that have been common across years. The High-Intensity Phenotyping Site (HIPS) experiment consisted of 22 hybrids (experiment called “HIP_Hybrid”) and 22 inbreds (experiment called “HIP_Inbred”).

Table 1. Description of the columns in the data file.

Name	Description
Year	Year of evaluation.
Field-Location	G2F Field-location name.
State	State name.
City	City name.
Plot Length	Plot length center to center in feet.
Plot Area	Calculated plot area in square feet from length and row spacing.
Alley Length	Length of alley in inches.
Row Spacing	Space between rows in inches.
Rows per plot	The number of rows per plot.
# Seed per plot	Total number of seed per plot.
Experiment	Experiment name
Pedigree	Pedigree name.
Family	Family grouping associated with hybrid.
Tester	Grouping association of the hybrid.
Replicate	Replicate block.
Block	Combination of Field-Location, Replicate, Tester and Family.
Plot	Designation of individual experimental unit.
Plot_ID	Unique designation of individual experimental unit considering all experiments.
Range	Designation of field range of the plot (ranges are organized perpendicular to corn rows).
Pass	Designation of field pass of the plot (passes are organized parallel to corn rows. Combination of range and pass form coordinate grid system describing location of each plot within the field).
Date Plot Planted	Date the plot was planted. (Month/Day/Year)
Date Plot Harvested	Date the plot was harvested. (Month/Day/Year)
Anthesis [date]	Date that 50% of plants, in the plot, began shedding pollen. (Month/Day/Year)
Silking [date]	Date that 50% of plants, in the plot, had visible silks. (Month/Day/Year)
Anthesis [days]	Number of days after planting that 50% of plants, in the plot, began shedding pollen.
Silking [days]	Number of days after planting that 50% of plants, in the plot, had visible silks.
Plant Height	Measured as the distance between the base of a plant and the ligule of the flag leaf (centimeter).
Ear Height	Measured as the distance from the ground to the primary ear bearing node (centimeter).
Stand count	Number of plants, per plot, at harvest.
Root Lodging	Number of plants, per plot, that show root lodging.
Stalk lodging	Number of plants, per plot, broken between ground level and top ear node at harvest.
Grain Moisture	Water content in grain at harvest (percentage).
Test Weight	Shelled grain weight per bushel (lbs/bu).
Plot Weight	Shelled grain weight per plot (lbs).
Grain yield*	Grain yield in bushels per acre at 15.5% grain moisture assuming 56lbs per bushel and using plot area without alley (bu/acre).
Plot Discarded	Value 'yes' if investigator believes no data from the plot should be used due to problem affecting all data.
Comments	Plot-specific comments provided by the investigator.
Filler	Pedigree name or 'yes' if a filler was used as a substitute for the original entry.
Snap	Green Snap: Number of plants broken between ground level and top ear node before flowering.

Grain yield = (Plot Weight(lbs)/plot area(ft^2))((100- Grain Moisture)/84.5))*(1(bu)/56(lbs))*(43560(ft^2)/1(acre))

Table 2. Conditions for declaring observations out of range.

Trait	Condition for setting to missing	Columns set to missing
Anthesis [days]	Less than 20 or more than 100 days[+]	Anthesis [days]
Silking [days]	Less than 20 or more than 100 days[+]	Silking [days]
Plant height	Less than 20 cm	Plant height
Ear height	Less than 20 cm	Ear height
Plot weight	Less than 1 lb	Plot weight, Grain yield, Grain moisture, Test weight
Grain yield	Less or equal to 0, or more than 300 bu/acre	Plot weight, Grain yield, Grain moisture, Test weight
Grain moisture	Less or equal to 0, or more than 45%	Plot weight, Grain yield, Grain moisture, Test weight
Test weight	Less or equal to 0	Plot weight, Grain yield, Grain moisture, Test weight
Stand count	Equal to 0	All traits

[+] Except for GEH1.